

Modeling the Spin Evolution of SMC X-1: Theoretical Framework and Calculations

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Abstract

This document presents the theoretical framework for modeling the spin period evolution of the high-mass X-ray binary SMC X-1. The calculations can be also applied to other objects. We combine observational constraints from Kahabka and Li (1999) with the accretion torque model of Dai and Li (2006) to investigate the relationship between magnetic moment, inner disk radius, and spin period derivative. The accompanying Python code implements these calculations to explore how different magnetic field strengths affect the accretion regime and predicted spin evolution.

1 Introduction

Observations of SMC X-1 by Kahabka and Li (1999) revealed a steady spin-up rate of $\dot{P} = -1.18 \times 10^{-11} \text{ s s}^{-1}$ over 6.5 years, with no observed spin-down episodes. I want to put a constraint on the pulsar's magnetic field strength and accretion dynamics, and reproduce the calculations vaguely presented in Kahabka and Li (1999).

2 Theoretical Framework

2.1 Basic Parameters and Assumptions

We adopt the following standard neutron star parameters based on typical values in the literature:

- Neutron star mass: $M = 1.4M_{\odot} = 2.78 \times 10^{33} \text{ g}$
- Moment of inertia: $I = 10^{45} \text{ g cm}^2$
- Pulse period: $P = 0.71 \text{ s}$ (from Kahabka and Li (1999))
- Accretion rate: Derived from X-ray luminosity $L_x \sim 5 \times 10^{38} \text{ erg s}^{-1}$, assuming 10% radiative efficiency:

$$\dot{M} = \frac{L_x}{\eta c^2} \approx 5.56 \times 10^{18} \text{ g s}^{-1}$$

2.2 Corotation Radius

The corotation radius R_c is defined as the radius where the Keplerian orbital period equals the neutron star spin period:

$$R_c = \left(\frac{GMP^2}{4\pi^2} \right)^{1/3} \quad (1)$$

where $G = 6.67430 \times 10^{-8} \text{ cm}^3 \text{ g}^{-1} \text{ s}^{-2}$ is the gravitational constant. For our parameters, $R_c \approx 1.5 \times 10^8 \text{ cm}$ ($\sim 150 \text{ km}$).

2.3 Fastness Parameter

The fastness parameter ω , a key dimensionless quantity introduced by ?, characterizes the relative rotation rate:

$$\omega = \frac{\Omega_*}{\Omega_K(R_{\text{in}})} = \left(\frac{R_{\text{in}}}{R_c} \right)^{3/2} \quad (2)$$

where $\Omega_* = 2\pi/P$ is the neutron star angular velocity and $\Omega_K(R) = \sqrt{GM/R^3}$ is the Keplerian angular velocity. The value of ω determines the accretion regime:

- $\omega < 1$: Accretion regime (material can accrete onto the star)
- $\omega > 1$: Propeller regime (centrifugal barrier prevents accretion)

2.4 Magnetospheric Radius

The magnetospheric radius R_m is the inner boundary of the accretion disk where magnetic pressure balances ram pressure. Following the standard formulation from Davidson and Ostriker (1973) and Lamb et al. (1973):

$$R_m = \left(\frac{\mu^4}{2GM\dot{M}^2} \right)^{1/7} \quad (3)$$

where $\mu = B_* R_*^3$ is the magnetic dipole moment. For a given $\dot{M}$, different μ values yield different R_m , which in turn determines ω through Equation (2).

2.5 Accretion Torque Model

We employ the modified accretion torque model from Dai and Li (2006), which addresses limitations of the classical Ghosh and Lamb (1979) model. The total torque N exerted on the neutron star consists of two components:

$$N = N_0 + N_{\text{mag}} \quad (4)$$

2.5.1 Material Torque N_0

The material torque represents angular momentum transfer by accreting matter:

$$N_0 = \xi \dot{M} \sqrt{GM R_{\text{in}}} (1 - \omega) \quad (5)$$

where $\xi \approx 1$ is an efficiency parameter accounting for angular momentum transfer in the boundary layer between the magnetosphere and disk.

2.5.2 Magnetic Torque N_{mag}

The magnetic torque arises from shearing of magnetic field lines between the neutron star and accretion disk. Following Dai and Li (2006):

$$N_{\text{mag}} = \begin{cases} \frac{\sqrt{2}\gamma}{3} \dot{M} \sqrt{GM R_{\text{in}}} \left(1 - 2\omega + \frac{2}{3}\omega^2\right), & \omega \leq 1 \\ \frac{\sqrt{2}\gamma}{3} \dot{M} \sqrt{GM R_{\text{in}}} \left(\frac{2}{3\omega} - 1\right), & \omega > 1 \end{cases} \quad (6)$$

where $\gamma \approx 1$ is a parameter characterizing the azimuthal pitch of the magnetic field. This formulation allows for continuous torque variation across the $\omega = 1$ boundary, addressing the singularity problem in classical models.

2.6 Spin Period Derivative

The spin period evolution is determined by the torque through:

$$-2\pi I \frac{\dot{P}}{P^2} = N \quad \Rightarrow \quad \dot{P} = -\frac{P^2 N}{2\pi I} \quad (7)$$

A negative $\dot{P}$ indicates spin-up (period decreasing), while a positive $\dot{P}$ indicates spin-down (period increasing). The observed value for SMC X-1 is $\dot{P} = -1.18 \times 10^{-11} \text{ s s}^{-1}$ (Kahabka and Li, 1999).

3 Computational Implementation

3.1 Parameter Space Exploration

The Python function `calculate_pdot_vs_rin_with_mu` calculates the relationship between inner disk radius (normalized to R_c) and the resulting $\dot{P}$, while also calculating the magnetic moment μ required for each R_{in} to be the magnetospheric radius for a given $\dot{M}$.

3.2 Key Calculations

Following are the default values of the parameters of the function. They can be changed by setting the corresponding arguments.

1. **Corotation radius:** Calculated from Equation (1)
2. **Inner radius range:** R_{in}/R_c from 0.01 to 1.1 (logarithmic spacing)
3. **Fastness parameter:** $\omega = (R_{\text{in}}/R_c)^{3/2}$
4. **Required magnetic moment:** Inverted from Equation (3):

$$\mu = \left(2GM\dot{M}^2 R_{\text{in}}^7\right)^{1/4}$$

5. **Torque calculation:** Using Equations (5) and (6)
6. **Spin period derivative:** From Equation (7)

3.3 Interpretation of Results

The critical insight from this analysis is that for a fixed accretion rate $\dot{M}$, the system naturally settles at R_m determined by μ . Therefore:

- Different μ values correspond to different positions on the R_{in}/R_c axis
- The observed steady spin-up of SMC X-1 requires $\omega < 1$ (accretion regime)

This supports the argument by Kahabka and Li (1999) that SMC X-1 likely has a relatively low magnetic moment, unlike systems like Cen X-3 that show torque reversals.

4 Discussion and Implications

4.1 Comparison with Classical Models

The Dai and Li (2006) model differs from classical accretion torque models in several key aspects:

- Allows accretion to continue in the “soft” propeller regime ($\omega \gtrsim 1$)
- Eliminates the torque singularity at $\omega = 1$
- Predicts spin-down torque that can increase with accretion rate
- Critical fastness parameter $\omega_c \approx 0.884$ where torque changes sign

4.2 Constraints on SMC X-1 Magnetic Field

For SMC X-1's observed steady spin-up to be consistent with the torque model (see Fig. 1 and 2):

1. If $\mu \sim 10^{30}$ G cm³: $R_m \sim R_c$ ($\omega \sim 1$), placing the system near the accretion/propeller boundary. A small decrease in $\dot{M}$ could push $\omega > 1$, leading to spin-down, which contradicts observations.
2. If $\mu < 10^{29}$ G cm³: $R_m \ll R_c$ ($\omega \ll 1$), keeping the system firmly in the accretion regime. Even with $\dot{M}$ variations, the system remains in spin-up, consistent with observations.

This analysis supports the conclusion by Kahabka and Li (1999) that SMC X-1 has a magnetic moment $\mu \sim 10^{29}$ G cm³, similar to the bursting pulsar GRO J1744-28.

References

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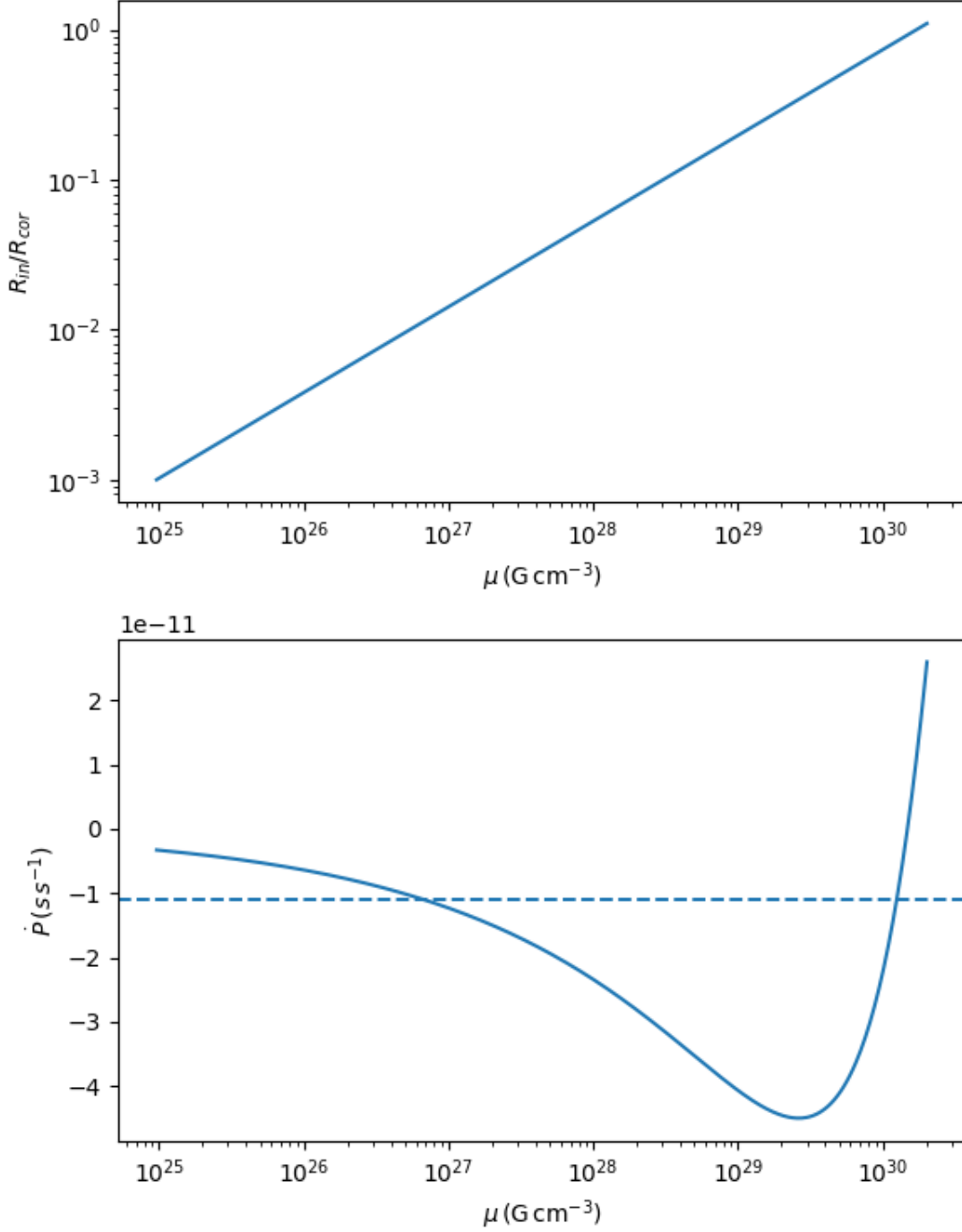


Figure 1: The output of the Python function `calculate_pdot_vs_rin_with_mu`. The horizontal dashed line represents the measured $\dot{P}$ of SMC X-1 in Kahabka and Li (1999). It has two solutions corresponding to 1. a weak μ and a high R_{in} ; or 2. a high μ and $R_{\text{in}} \approx R_{\text{cor}}$.

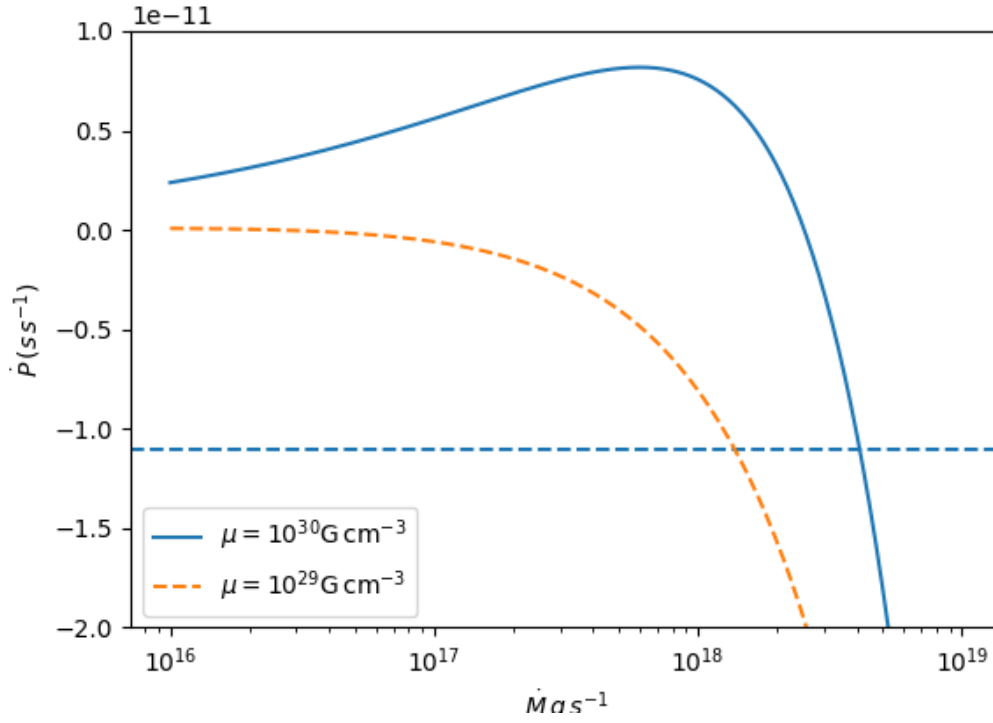


Figure 2: The output of the Python function `calculate_pdot_vs_mdots`. The horizontal dashed line represents the measured $\dot{P}$ of SMC X-1 in Kahabka and Li (1999). If the mass accretion rate decreases by a small factor and $\mu = 10^{30} \text{G cm}^{-3}$, SMC X-1 would have spun down. Therefore it is more likely that $\mu \leq 10^{29} \text{G cm}^{-3}$.